CCS4711 CRYPTOGRAPHY AND APPLICATIONS

SEMESTER DEFERRED EXAMINATION

THURSDAY,19 JUNE 2025

09:00 – 12:00 HOURS

TIME ALLOWED : WRITING - THREE HOURS : READING — 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is compulsory and must be attempted.
2. Sections B: Answer THREE (3) questions from this section.
3. This examination paper carries a total of 100 marks.
4. Candidates must not turn this page until the invigilator tells them to do so.

Question 1 – Compulsory

a) Using a clearly-labelled diagram, describe how **Data Encryption Standard XORed (DESX)** extends classic DES. Explain the purpose of pre-whitening and post-whitening keys and analyse how DESX increases exhaustive-key-search complexity. (10 marks)

b) Outline the **Encrypt-Decrypt-Encrypt (EDE)** workflow of 3DES. Compare the effective key lengths of the *two-key* and *three-key* variants, and discuss one real-world application where 3DES is still preferred over AES today. (10 marks)

c) Give a general proof showing why a Feistel structure can use the **same round function for both encryption and decryption**. Illustrate your answer with a 4-round example, clearly indicating the data flow in each direction. (10 marks)

d) For each of the following modes—**ECB, CBC, OFB**—explain:
i. How the next input to the cipher is formed.
ii. Whether blocks can be processed in parallel.
iii. What happens when a single ciphertext bit is flipped in transit.
Provide a concise use-case where each mode is appropriate. (10 marks)

SECTION B (Answer ANY THREE questions – 60 Marks)

Question 2 (20 Marks)

a) Contrast the El Gamal digital signature scheme with El Gamal encryption in each of the following aspects, supplying one concise example for every point:

i. Cryptographic goal (what security property each scheme provides)

ii. Key-pair structure and parameter requirements

iii. Typical use-case scenario in which one would choose the signature scheme over the encryption scheme (or vice-versa) (8 marks)

b) A ciphertext $C = 8\,237$ is received under the public key $(n, e) = (3\,713, 17)$.
Demonstrate step-by-step **CRT-based decryption**, clearly highlighting speed-ups gained versus direct modular exponentiation. (8 marks)

c) State **four reasons** why modern RSA deployments wrap messages in padding schemes such as **OAEP** rather than encrypting raw plaintext blocks. (4 marks)

Question 3 (20 Marks)

a) Define the **Discrete Logarithm Problem (DLP)** and explain why its presumed hardness underpins the security of Diffie-Hellman key exchange. (6 marks)

b) Given prime $p = 467$ and generator $g = 2$, Alice picks private key $a = 91$ while Bob chooses $b = 233$.
i. Compute the public values each participant sends.
ii. Derive the shared secret they both obtain.
Show calculations. (7 marks)

c) Compare **Classic Diffie-Hellman** over prime fields with **Elliptic Curve Diffie-Hellman (ECDH)** in terms of key size, computational cost, and resistance to quantum-computing threats. Provide one practical example where ECDH offers a decisive advantage. (7 marks)

Question 4 (20 Marks)

a) Summarise the main steps in generating and verifying an **Elliptic Curve Digital Signature Algorithm (ECDSA)** signature, noting the role of the deterministic nonce *k*. (6 marks)

b) Compare **software** and **hardware** implementations of RSA with respect to each of the following aspects. Provide a concise explanation and one practical example in each case.

i. Resistance to side-channel attacks
ii. Throughput when generating 2048-bit digital signatures
iii. Typical real-world deployment scenarios (e.g., smartcards, HSM-backed key

(7 marks)

c) List **three reasons** ECDSA is favoured over RSA for securing firmware updates on low-power Internet-of-Things (IoT) devices. (7 marks)

Question 5 (20 Marks)

a) Describe *soft*, *hard*, and *accidental* blockchain forks, including one technical scenario that can trigger each type. (10 marks)

c) Trace the life-cycle of a cryptocurrency transaction from creation to confirmation, focusing on how **cryptographic hash functions** and **digital signatures** together guarantee authenticity, integrity, and non-repudiation. (10 marks)

END OF EXAMINATION